# Oblique Feshbach Determinant Factorization

## Exact Packet–Complement Coordinates for a Biorthogonal Temporal Clock

Liang Wang

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**Abstract**

RH-188 identifies a surviving directional coupling product but leaves the packet and complement resolvents undefined. This paper supplies the exact finite block decomposition for a biorthogonal temporal packet.

Let $V, W \in \mathbb{C}^{n\times r}$ satisfy $W^*V = I_r$. Choose a frame $Z$ for $\ker W^*$ and set $S = [V, Z]$. Then $S$ is invertible and its inverse has the form $S^{-1} = [W^*; Y^*]$, with $Y^*V = 0$ and $Y^*Z = I$. In these coordinates

$$S^{-1}AS = \begin{pmatrix} K & B \\ C & D \end{pmatrix}, \quad K = W^*AV.$$

Whenever $zI - D$ is invertible,

$$\det(zI - A) = \det(zI - D)\det\Big(zI - K - B(zI - D)^{-1}C\Big).$$

This is a type-correct oblique Feshbach factorization. The self-energy is the exact object controlled by the directed Schur product.

A 240-case complex random audit verifies coordinate inversion, block similarity, complement-gauge invariance, the one-factor norm bound, and the determinant identity with zero failures. The maximum relative determinant error is $1.85 \times 10^{-11}$. The theorem converts the local bi-Krylov candidate into a precise next question: validate the complement resolvent on root contours. It does not itself supply those bounds or prove a physical Riesz shell.

# 1 Why an exact coordinate theorem is needed

The RH-185 packet is not orthogonal. Writing $Q = I - VW^*$ and informally calling $QAQ$ the complement block hides two issues: $Q$ is oblique, and an operator on $\operatorname{Ran} Q$ needs a chosen coordinate norm. A determinant identity should be derived from an actual similarity, not from a symbolic orthogonal block matrix.

Finite-dimensional Feshbach and Schur complement identities are classical [1]. The contribution here is to state them in the exact packet type produced by RH-184–185 and to make the complement coordinates explicit.

## 2 Biorthogonal packet coordinates

Let $A \in \mathbb{C}^{n\times n}$ and let $V, W \in \mathbb{C}^{n\times r}$ satisfy

$$W^*V = I_r. \tag{1}$$

The associated oblique projector is $P = VW^*$.

Choose any full-rank

$$Z : \mathbb{C}^{n-r} \longrightarrow \ker W^*. \tag{2}$$

**Proposition 2.1** (Invertible packet–complement coordinates)**.** *The matrix*

$$S = [V, Z] \tag{3}$$

*is invertible. Its inverse has a unique block-row representation*

$$S^{-1} = \begin{bmatrix} W^* \\ Y^* \end{bmatrix} \tag{4}$$

*with*

$$Y^*V = 0, \qquad Y^*Z = I_{n-r}. \tag{5}$$

*Proof.* Suppose $Va + Zb = 0$. Applying $W^*$ and using $W^*V = I$, $W^*Z = 0$ gives $a = 0$. Then $Zb = 0$, hence $b = 0$ because $Z$ has full rank. Thus $S$ is invertible. Multiplying $S^{-1}S = I$ shows that its top row must be $W^*$ and its bottom row satisfies (5). □

The choice of $Z$ is a complement-coordinate gauge. Different choices are related by an invertible block-triangular similarity and do not change the full determinant.

For norm estimates there is a canonical convenient choice.

**Proposition 2.2** (Orthonormal complement coordinates)**.** *Let $Z$ have orthonormal columns spanning* $\ker W^*$ *and put $Q = I - VW^*$. Then the lower block row of $S^{-1}$ is*

$$Y^* = Z^*Q. \tag{6}$$

*Consequently*

$$D = Z^*QAZ, \qquad \|D\| \le \|Q\| \, \|A\| \, . \tag{7}$$

*Proof.* Because $QZ = Z$ and $QV = 0$, the row $Z^*Q$ satisfies $(Z^*Q)V = 0$ and $(Z^*Q)Z = I$. Uniqueness in Proposition 2.1 gives (6). The block formula and its norm bound follow immediately. □

This removes one unnecessary oblique factor from the elementary complement bound. The ambient operator $QAQ$ obeys the valid but looser estimate $\|QAQ\| \le \|Q\|^2 \|A\|$; the actual Feshbach coordinate block in an orthonormal complement needs only one factor $\|Q\|$.

# 3 Exact block operator

Define

$$S^{-1}AS = \begin{pmatrix} K & B \\ C & D \end{pmatrix}, \tag{8}$$

where

$$K = W^*AV, \qquad B = W^*AZ, \tag{9}$$
$$C = Y^*AV, \qquad D = Y^*AZ. \tag{10}$$

The packet block $K$ is exactly the RH-185 compression. The off-diagonal blocks are coordinate forms of the RH-188 directed couplings.

**Proposition 3.1** (Residual interpretation)**.** *The ambient right residual satisfies*

$$AV - VK = ZC, \tag{11}$$

*and the left residual satisfies*

$$A^*W - WK^* = YB^*. \tag{12}$$

*Proof.* From $AS = S(\begin{smallmatrix} K & B \\ C & D \end{smallmatrix})$, the first block column gives $AV = VK + ZC$. Taking the adjoint of the top block row of $S^{-1}A = (\begin{smallmatrix} K & B \\ C & D \end{smallmatrix})S^{-1}$ gives the second identity. □

Thus the numerical residuals and the Feshbach couplings are the same data in different coordinate norms.

If $Z$ is replaced by $Z' = ZG$ for an invertible complement gauge $G$, then $Y'^* = G^{-1}Y^*$ and

$$B' = BG, \qquad C' = G^{-1}C, \qquad D' = G^{-1}DG. \tag{13}$$

It follows that

$$B'(zI - D')^{-1}C' = B(zI - D)^{-1}C. \tag{14}$$

Thus the self-energy is independent of the complement basis, even though the three separate block norms depend on that basis.

# 4 Feshbach determinant identity

For a spectral parameter $z$ with $zI - D$ invertible, define the self-energy

$$\Sigma(z) = B(zI - D)^{-1}C \tag{15}$$

and the reduced Feshbach matrix

$$F(z) = zI - K - \Sigma(z). \tag{16}$$

**Theorem 4.1** (Oblique determinant factorization)**.** *Whenever $zI - D$ is invertible,*

$$\boxed{\det(zI - A) = \det(zI - D)\det F(z).} \tag{17}$$

*Proof.* Similarity by $S$ gives

$$\det(zI - A) = \det \begin{pmatrix} zI - K & -B \\ -C & zI - D \end{pmatrix}.$$

Multiply on the left by the determinant-one block matrix

$$\begin{pmatrix} I & B(zI - D)^{-1} \\ 0 & I \end{pmatrix}.$$

The result is block lower triangular with diagonal blocks $F(z)$ and $zI - D$. Taking determinants proves (17). □

**Corollary 4.2** (Schur sufficient condition)**.** *If $zI - K$ and $zI - D$ are invertible and*

$$\left\|(zI - K)^{-1}\right\| \|B\| \left\|(zI - D)^{-1}\right\| \|C\| < 1, \tag{18}$$

*then $F(z)$ is invertible and $z \notin \sigma(A)$.*

*Proof.* Factor $F(z) = (zI - K)[I - (zI - K)^{-1}\Sigma(z)]$ and apply the Neumann lemma. □

This is the exact location of the four factors recorded architecturally in RH-163 and numerically separated in RH-188.

# 5 Riesz rank consequence

The determinant identity also gives the precise counting statement. The complement count cannot be silently omitted.

**Theorem 5.1** (Riesz count under a uniform Schur margin)**.** *Let $\Gamma$ be a positively oriented simple contour on which both $zI - K$ and $zI - D$ are invertible. Suppose*

$$\sup_{z\in\Gamma} \left\|(zI - K)^{-1}\right\| \|B\| \left\|(zI - D)^{-1}\right\| \|C\| < 1. \tag{19}$$

*Then the algebraic eigenvalue count satisfies*

$$N_A(\Gamma) = N_K(\Gamma) + N_D(\Gamma). \tag{20}$$

*Here $N_T(\Gamma)$ denotes the algebraic multiplicity of the spectrum of $T$ inside $\Gamma$. In particular, $N_A(\Gamma) = N_K(\Gamma)$ if the complement block has no eigenvalues inside $\Gamma$.*

*Proof.* For $0 \le t \le 1$, define

$$A_t = S \begin{pmatrix} K & tB \\ tC & D \end{pmatrix} S^{-1}.$$

Its reduced matrix is $F_t(z) = zI - K - t^2 B(zI - D)^{-1}C$. The bound (19) makes $F_t(z)$ invertible on $\Gamma$ for every $t$. The full determinant therefore has no contour zero during the homotopy, so its winding number is constant. At $t = 0$ the operator is block diagonal and its count is $N_K + N_D$. □

The theorem is conditional on continuous packet and complement inverse bounds. To isolate one packet root, one must either validate $N_D(\Gamma) = 0$ or compute and retain the complement contribution. Small residuals alone imply neither fact.

# 6 Finite identity audit

The audit uses dimensions $5, 7, 9, 11$, packet ranks $1, 2, 3$, and twenty complex random trials per pair, for 240 cases. The left frame is corrected to satisfy $W^*V = I$ exactly at floating precision. A numerical nullspace of $W^*$ supplies $Z$, and $Y$ is read from $S^{-1}$.

The maximum errors are:

| identity | maximum error |
|---|---|
| $S^{-1}S = I$ | $3.22 \times 10^{-14}$ |
| block similarity | $1.01 \times 10^{-11}$ |
| relative determinant factorization | $1.85 \times 10^{-11}$ |
| orthonormal complement dual row | $2.38 \times 10^{-12}$ |
| one-factor norm-bound violation | 0 |
| complement-gauge self-energy | $2.64 \times 10^{-12}$ |

All 240 cases pass the declared $10^{-8}$ tolerance.

The identity audit includes nominal recomputation of the self-energy in a second complement gauge. A later validated implementation must additionally enclose that equality. This catches a numerically important failure mode: separate block norms can change greatly under a poor complement basis even though the exact reduced determinant is fixed.

For outward work, the orthonormal choice in Proposition 2.2 is the baseline because it gives the one-factor bound $\|D\| \le \|Q\| \, \|A\|$. Another gauge is useful only if the transformation and its improved product bound are both validated.

# 7 Physical next step

For each surviving RH-185 window, the theorem defines a finite exact program:

1. build outward enclosures of $V, W, K, B, C, D$;
2. choose root contours from the predeclared length-four geometry;
3. validate $(zI - D)^{-1}$ on a finite contour mesh;
4. combine mesh, operator-ball, and coupling errors using the RH-167–168 machinery summarized in RH-171 [2];
5. apply (18), compute $N_D$, and use Theorem 5.1 to count the full Riesz rank.

The only genuinely new hard object is the physical complement resolvent. The packet and coupling types are now exact.

# 8 Boundary

This paper proves a finite determinant identity for every biorthogonal packet pair. It does not validate any physical complement inverse, establish a continuous Schur margin, prove a physical Riesz shell, transport shells across scales, close R or Gate A, or approach a self-adjoint Hilbert–Polya operator or the Riemann Hypothesis.

# References

[1] Israel Gohberg, Seymour Goldberg, and Marinus A. Kaashoek. *Classes of Linear Operators, Volume I.* Birkhäuser, 1990.

[2] Liang Wang. Ten layers toward the physical riesz interface, 2026. Prime Dynamics Theory Program, RH-171.